DA5400 – Foundations of Machine Learning

Regression Tutorial

Questions 1 · 2 · 7

Q1 Linearizable Models

Q2 OLS Regression

Q7 OLS Derivation

QUESTION STATEMENT

Question 1 — Linearizable Regression Models

Which of the following equations can be reformulated as linear regression models, treating (x, y) as variables and (a, b) as parameters?

For each valid case, show the algebraic transformation that converts the equation into linear regression form.

(a) $y = a \cdot e^{bx}$

(b) $y = ax / (b + x)$

(c) $y = a / (x + b)$

(d) $y = a(1 - e^{-bx})$

Question 1 — Linearizable Regression Models

Can we transform nonlinear equations into linear regression form?

💡 **Core Idea**

A model is linearizable if a mathematical transformation (log, reciprocal, etc.) converts it into the form $\tilde{y} = a \cdot \tilde{x} + b$, where $\tilde{y}$ and $\tilde{x}$ are transformed variables but a, b remain LINEAR parameters.

Original form

$y = f(a, b, x)$ — nonlinear

Goal

Find T_1, T_2 so that $T_1(y) = a \cdot T_2(x) + b$

Key check

Parameters a, b must appear linearly after transformation

Linear Regression Form

$$\tilde{y} = a \cdot \tilde{x} + b$$

where $\tilde{y}$ = transformed output

$\tilde{x}$ = transformed input

a, b = parameters (linear in model)

Four sub-cases follow on next slides ➜

Q1(a) — $y = a \cdot e^{bx}$

Exponential model → linearizable via natural logarithm

1 Start with the original model

$$y = a \cdot e^{bx}$$

Exponential in x — clearly nonlinear

2 Apply ln (natural log) to both sides

$$\ln y = \ln(a \cdot e^{bx}) = \ln a + bx$$

Using: $\ln(PQ) = \ln P + \ln Q$ and $\ln(e^k) = k$

3 Substitute transformed variables

$$\tilde{y} = \ln y, \quad A = \ln a, \quad \text{then } \tilde{y} = A + b \cdot x$$

Now $\tilde{y}$ is linear in x with slope b and intercept $A = \ln a$

4 Linear regression form

$$\tilde{y} = b \cdot x + A$$

Recover: $a = e^A$ after fitting

Summary: Take ln of y. Plot ln y vs x. Fit a line → slope = b, intercept = A = ln a → $a = e^A$

Q1(b) — $y = ax / (b + x)$

Michaelis-Menten / saturation model → linearizable via reciprocal

1 Start with the model

$$y = ax / (b + x)$$

Nonlinear — cannot be written directly as a linear combination of parameters

2 Take reciprocal of both sides

$$1/y = (b + x) / (ax) = b/(ax) + x/(ax) = b/(ax) + 1/a$$

Rewrite right-hand side by splitting the fraction

3 Rearrange into linear form

$$1/y = (b/a) \cdot (1/x) + 1/a$$

Intercept = 1/a, Slope = b/a → both are linear in 1/a and b/a

4 Substitute $\tilde{y} = 1/y$, $\tilde{x} = 1/x$

$$\tilde{y} = (b/a) \cdot \tilde{x} + (1/a)$$

Standard linear form. Recover: a = 1/intercept, b = slope / intercept

Q1(c) — $y = a / (x + b)$

Rational model — is it linearizable?

1 Take reciprocal: $1/y = (x + b)/a$

$$1/y = x/a + b/a$$

Looks linear in x — but with parameters 1/a and b/a

2 Substitute $\tilde{y} = 1/y$

$$\tilde{y} = (1/a) \cdot x + (b/a)$$

slope = 1/a, intercept = b/a → BOTH are linear in the transformed parameters

3 Conclusion — actually IS linearizable!

$$\tilde{y} = (1/a) \cdot x + b/a \text{ , i.e., plot } 1/y \text{ vs } x$$

Recover: a = 1/slope, b = intercept × a = intercept/slope

⚠ Correction: $y = a/(x+b)$ IS linearizable. Taking $1/y$ gives $1/y = (1/a)\cdot x + b/a$ — a line in x.
Slope = 1/a → a = 1/slope. Intercept = b/a → b = intercept · a.

Q1(d) — $y = a(1 - e^{-bx})$

Growth saturation model — is it linearizable?

1 **Attempt: isolate the exponential term**

$$y/a = 1 - e^{-bx} \implies e^{-bx} = 1 - y/a$$

Requires knowledge of a before we can proceed — a appears nonlinearly

2 **Take ln of both sides**

$$-bx = \ln(1 - y/a) \implies \ln(1 - y/a) = -b \cdot x$$

This form has BOTH a and b as unknowns inside the log — not linear in a

3 **Why this fails the linearity test**

$$\partial^2\tilde{y} / \partial a^2 \neq 0 \quad \text{(a appears inside ln — nonlinear parameter)}$$

No algebraic rearrangement makes both a and b appear linearly simultaneously

Verdict: $y = a(1 - e^{-bx})$ is NOT linearizable.

Q1 Summary: (a) ✔ ln y vs x | (b) ✔ 1/y vs 1/x | (c) ✔ 1/y vs x | (d) ✘ Not linearizable

QUESTION STATEMENT

Question 2 — OLS Simple Linear Regression from Summary Statistics

Data on the monthly sales (y) in rupees and advertising spend (x) in rupees for 25 retail stores were collected. The mean and covariance matrix (about the mean) computed from the data is given:

$$\mathbf{z} = [y, x]^T, \quad \bar{\mathbf{z}} = [200, 40]^T, \quad \mathbf{S}_u = [[1800, 900], [900, 500]]$$

Assume a linear relation $y = ax + b$ exists between monthly sales (y) and advertising spend (x). Compute the model parameters using the OLS approach.

Question 2 — OLS Simple Linear Regression

Monthly sales (y) vs advertising spend (x) for 25 retail stores

💡 We are given summary statistics (mean + covariance matrix) — NOT raw data. OLS formulas can be expressed entirely in terms of these statistics.

Given Data

$n = 25$ stores

$\bar{y} = 200$ (mean monthly sales, ₹)

$\bar{x} = 40$ (mean ad spend, ₹)

$Var(y) = S_yy = 1800$

$Var(x) = S_xx = 500$

$Cov(x,y) = S_xy = 900$

Model to Fit

$$y = a \cdot x + b$$

where a = slope, b = intercept

OLS Estimator Formulas

$\hat{a} = S_xy / S_xx = Cov(x,y) / Var(x)$

$\hat{b} = \bar{y} - \hat{a} \cdot \bar{x}$

(OLS line always passes through $(\bar{x}, \bar{y})$)

Step-by-step solution ➜ next slide

Q2 — Step-by-Step OLS Solution

Computing slope $\hat{a}$ and intercept $\hat{b}$ from summary statistics

1 Extract values from the covariance matrix

$$S_xx = Var(x) = 500, \quad S_xy = Cov(x,y) = 900$$

Read directly from the given matrix Sz

2 Compute the OLS slope $\hat{a}$

$$\hat{a} = S_xy / S_xx = 900 / 500 = 1.8$$

Slope = 1.8 → for each additional ₹1 in ad spend, sales increase by ₹1.8

3 Compute the OLS intercept $\hat{b}$

$$\hat{b} = \bar{y} - \hat{a} \cdot \bar{x} = 200 - 1.8 \times 40 = 200 - 72 = 128$$

Intercept = 128 → baseline sales when ad spend = 0

4 Write the fitted model

$$\hat{y} = 1.8x + 128$$

This is our OLS linear regression model for the 25-store dataset

Final Answer: $\hat{a} = 1.8$ (slope) $\hat{b} = 128$ (intercept) → $\hat{y} = 1.8x + 128$

Q2 — Interpretation & Verification

Understanding what the OLS estimates tell us

✔ Verification: OLS line always passes through $(\bar{x}, \bar{y})$

Check: $\hat{y}$ at $x = \bar{x} = 40$: $\hat{y} = 1.8 \times 40 + 128 = 72 + 128 = 200 = \bar{y}$ ✔

Slope $\hat{a} = 1.8$

For every additional ₹1 spent on advertising, expected monthly sales increase by ₹1.80. This is the marginal return on ad spend.

Intercept $\hat{b} = 128$

Baseline monthly sales when ad spend = ₹0 is ₹128. This represents the inherent sales without any advertising.

OLS Key Property

OLS minimizes the sum of squared vertical distances (residuals) between observed y_i and predicted $\hat{y}_i = \hat{a}\, x_i + \hat{b}$.

Fitted model: $\hat{y} = 1.8x + 128$ | Correlation $r = S_{xy} / \sqrt{(S_{xx} \cdot S_{yy})} = 900 / \sqrt{(500 \times 1800)} = 900/948.7 \approx 0.949$

QUESTION STATEMENT

Question 7 — Derivation of OLS Estimators for Simple Linear Regression

Consider the simple linear regression model:

$$y = w_0 + w_1 x$$

Given N samples of data, the Ordinary Least Squares (OLS) estimator minimizes the

sum of squared residuals:

$$SSR = \Sigma_{i=1}^{N} (y_i - w_0 - w_1 x_i)^2$$

to estimate the parameters. Specifically:

Part (a) Derive the first-order conditions by differentiating SSR with respect to w_0 and w_1, and setting the derivatives equal to zero.

Part (b) Using the result from part (a), show that the OLS estimator of the slope is: $\hat{w}_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$

Question 7 — Deriving OLS Estimators

Formal derivation of slope and intercept from first principles

💡 Goal: Derive closed-form expressions for w_0 (intercept) and w_1 (slope) by minimizing the Sum of Squared Residuals (SSR).

Model

$$y = w_0 + w_1 x$$

w_0 = intercept, w_1 = slope

Objective: Minimize SSR

$$SSR = \Sigma_{i=1}^{N} (y_i - w_0 - w_1 x_i)^2$$

Sum of squared differences between observed and predicted y

a Part (a): Differentiate SSR w.r.t. w_0 and w_1, set = 0 (Normal Equations)

b Part (b): Solve Normal Equations to derive $\hat{w}_1$ in terms of sample means and deviations

Full derivation follows on the next two slides ➜

Q7(a) — First-Order Conditions (Normal Equations)

Differentiating SSR w.r.t. w_0 and w_1 and setting derivatives to zero

Step 1 — Differentiate SSR w.r.t. w_0

$$\partial SSR / \partial w_0 = \Sigma_i\, 2(y_i - w_0 - w_1 x_i)(-1) = 0$$

Divide by −2: $\Sigma_i\, (y_i - w_0 - w_1 x_i) = 0$

==> $\Sigma y_i = N \cdot w_0 + w_1 \cdot \Sigma x_i$ --- > divide by N: $\bar{y} = w_0 + w_1 \cdot \bar{x}$ ...(I)

Step 2 — Differentiate SSR w.r.t. w_1

$$\partial SSR / \partial w_1 = \Sigma_i\, 2(y_i - w_0 - w_1 x_i)(-x_i) = 0$$

Divide by −2: $\Sigma_i\, x_i(y_i - w_0 - w_1 x_i) = 0$

==> $\Sigma x_i y_i = w_0 \cdot \Sigma x_i + w_1 \cdot \Sigma x_i^2$ ...(II)

Equations (I) and (II) are the NORMAL EQUATIONS — two equations, two unknowns w_0 and w_1

Q7(b) — Solving for the OLS Slope Estimator $\hat{w}_1$

Eliminating w_0 from the normal equations to get the slope formula

1 From Normal Equation (I): express w_0

$$\hat{w}_0 = \bar{y} - \hat{w}_1 \cdot \bar{x}$$

The OLS line passes through the sample mean point $(\bar{x}, \bar{y})$

2 Substitute $\hat{w}_0$ into Normal Equation (II)

$$\Sigma x_i y_i = (\bar{y} - \hat{w}_1 \bar{x}) \cdot \Sigma x_i + \hat{w}_1 \cdot \Sigma x_i^2$$

Expand: $\Sigma x_i y_i = \bar{y} \cdot \Sigma x_i - \hat{w}_1 \bar{x} \cdot \Sigma x_i + \hat{w}_1 \cdot \Sigma x_i^2$

3 Rearrange to isolate $\hat{w}_1$

$$\Sigma x_i y_i - \bar{y} \cdot \Sigma x_i = \hat{w}_1 (\Sigma x_i^2 - \bar{x} \cdot \Sigma x_i)$$

Using $\Sigma x_i = N \cdot \bar{x}$ and collecting $\hat{w}_1$ terms on the right

4 Use identity: $\Sigma(x_i - \bar{x})(y_i - \bar{y}) = \Sigma x_i y_i - N \cdot \bar{x} \cdot \bar{y}$

$$\hat{w}_1 = \Sigma_i (x_i - \bar{x})(y_i - \bar{y}) \ / \ \Sigma_i (x_i - \bar{x})^2$$

This is the standard OLS slope formula — identical to S_xy / S_xx in Q2

Proved: $\hat{w}_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$ and $\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$ ■

Q7 — Geometric Intuition & Proof Summary

Why these formulas make sense

Geometric Intuition

- SSR is a convex quadratic in (w_0, w_1) → unique global minimum
- Setting $\partial SSR/\partial w_0 = 0$ means: residuals $\Sigma e_i = 0$ (they sum to zero)
- Setting $\partial SSR/\partial w_1 = 0$ means: $\Sigma x_i e_i = 0$ (residuals orthogonal to x)
- These two conditions uniquely determine the OLS line
- Numerator $\Sigma(x_i-\bar{x})(y_i-\bar{y})$ = sample covariance × N
- Denominator $\Sigma(x_i-\bar{x})^2$ = sample variance × N
- So: $\hat{w}_1 = Cov(x,y) / Var(x)$ (same as Q2!)

OLS Estimators — Final Result

Slope:

$$\hat{w}_1 = \Sigma(x_i-\bar{x})(y_i-\bar{y}) / \Sigma(x_i-\bar{x})^2$$

Equivalently:

$$\hat{w}_1 = S_xy / S_xx$$

Intercept:

$$\hat{w}_0 = \bar{y} - \hat{w}_1 \cdot \bar{x}$$

Normal Eq. 1:

$$\Sigma(y_i - \hat{y}_i) = 0$$

Normal Eq. 2:

$$\Sigma x_i(y_i - \hat{y}_i) = 0$$

Both Q2 and Q7 use the same formula — Q2 applies it numerically, Q7 derives it analytically.

Tutorial Summary — Q1, Q2, Q7

Q1 Linearizable Models

(a) $y=ae^{bx} \rightarrow \ln y = bx + \ln a$ ✔

(b) $y=ax/(b+x) \rightarrow 1/y = (b/a)(1/x) + 1/a$ ✔

(c) $y=a/(x+b) \rightarrow 1/y = (1/a)x + b/a$ ✔

(d) $y=a(1-e^{-bx}) \rightarrow$ NOT linearizable ✘

Q2 OLS from Summary Stats

$\hat{a} = S_{xy} / S_{xx} = 900/500 = 1.8$

$\hat{b} = \bar{y} - \hat{a}\cdot\bar{x} = 200 - 72 = 128$

Fitted model: $\hat{y} = 1.8x + 128$

Verified: passes through ($\bar{x}=40$, $\bar{y}=200$) ✔

Q7 OLS Derivation

Minimize SSR → 2 normal equations

$\partial SSR/\partial w_0=0$ x $\bar{y} = w_0 + w_1\bar{x}$

$\partial SSR/\partial w_1=0 \implies \Sigma x_i y_i = w_0 \Sigma x_i + w_1 \Sigma x_i^2$

$\hat{w}_1 = \Sigma(x_i-\bar{x})(y_i-\bar{y})/\Sigma(x_i-\bar{x})^2 = Cov/Var$